

Recovering Strategic Phase Structure in Recurrent Multi-Agent Socio-Economic Systems

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Abstract—Socio-economic strategic systems such as trade conflicts, sanctions regimes, and retaliatory policy competition involve adaptive multi-agent interactions with delayed feedback, asymmetric roles, and regime-dependent dynamics. Evaluating learned behavior in such systems through reward metrics alone is insufficient because qualitatively distinct strategic equilibria may yield similar episodic rewards while differing substantially in their phase topology. We study whether recurrent multi-agent learning systems can recover the phase topology of a two-country leader-follower tariff conflict under function approximation, using phase-topology recovery as the primary evaluation criterion. We establish a tabular double Q-learning baseline as a reference phase topology for evaluating recurrent multi-agent architectures under identical economic dynamics. Naive recurrent deep Q-networks fail to recover this topology due to three systematic failure modes: reward-scale domination, credit assignment collapse across factorized instruments, and recurrent overwrite of sparse strategic signals. We then identify a set of architectural and training conditions that restore topology recovery, including factorized action heads, a tau-conditional stateless retaliation pathway, decomposed Bellman targets, and aligned temporal credit assignment, instantiated in the Factorized DRQN (FactorDRQN). The resulting system achieves 92.2% cell-level regime agreement (59/64 cells) with the tabular reference on the primary structural parameter grid, and 90.6% agreement under a second independent perturbation axis, with disagreements concentrated near the Transition–Escalation boundary at high retaliation capacity. This work contributes a phase-topology evaluation framework, a failure mode taxonomy for recurrent socio-economic MARL systems, and a set of architecture conditions associated with stable topology recovery in strategic multi-agent domains.

Index Terms—Multi-agent reinforcement learning; recurrent deep Q-networks; phase-topology evaluation; socio-economic conflict; factorized architecture; regime structure

I. INTRODUCTION

Strategic interactions between adaptive agents appear across a wide range of socio-economic systems, from trade policy and sanctions enforcement to retaliatory regulatory competition. These settings share a common structure: two or more agents with asymmetric roles and instruments interact repeatedly over time, with delayed and partially observable feedback, and the system settles into qualitatively distinct behavioral regimes

depending on structural parameters. Reinforcement learning offers a natural framework for studying such interactions, since agents must discover effective policies without access to a closed-form equilibrium.

A persistent limitation of existing RL approaches to strategic economic domains is their reliance on reward metrics for evaluation. Episodic reward measures whether an agent performs well on average, but does not capture whether the system has converged to a structurally coherent strategic regime. Two systems with similar mean returns may differ substantially in their phase topology: one may produce a clean separation between deterrence and escalation regimes across structural parameters, while the other produces a fragmented pattern that fails to reflect the underlying strategic geometry. For multi-agent systems deployed in policy-relevant domains, this distinction matters.

Stable phase structure emerges under tabular learning in leader-follower conflict environments, producing interpretable regime boundaries across structural parameter grids that serve as low-dimensional characterizations of equilibrium behavior. The question this paper addresses is whether recurrent multi-agent architectures, which are commonly used in partially observable settings, can recover this phase topology under function approximation.

We show that naive recurrent deep Q-networks fail to do so. Applied directly to a two-country leader-follower tariff conflict, standard DRQN architectures produce fragmented or structurally collapsed topologies: the Transition band disappears, regime boundaries fragment, or a single regime dominates across the parameter grid. The failure is not random but systematic, arising from three identifiable pathological interactions between recurrent function approximation and the structure of strategic socio-economic environments.

We diagnose these failure modes and identify the architectural conditions that address them. The central insight is that instruments with different temporal structures require separated learning pathways. Retaliation in leader-follower conflict is near-Markov in the current tariff level: it depends primarily on the immediate provocation rather than accumulated history.

In our environment, entangling the retaliation pathway with a shared recurrent state suppresses this threshold structure and degrades topology recovery. Separating it into a stateless tau-conditional branch, alongside decomposed Bellman targets per instrument, restores the credit assignment alignment associated with stable topology recovery. We instantiate these principles in the Factorized DRQN (FactorDRQN) and evaluate it against the tabular reference topology across an 8×8 structural parameter grid.

This paper makes four contributions:

- **Phase-topology evaluation framework.** We propose evaluating recurrent MARL systems by their recovery of reference phase topology rather than by episodic return, and operationalize this through cell-level regime agreement, majority-vote aggregation across seeds, and boundary geometry comparison.
- **Failure mode taxonomy.** We identify and characterize three systematic failure modes of naive recurrent deep Q-networks in strategic socio-economic environments: reward-scale domination, credit assignment collapse across factorized instruments, and recurrent overwrite of sparse strategic signals.
- **Architecture conditions.** We identify three architectural conditions associated with stable topology recovery in this class of environments: strategic pathway factorization, tau-conditional stateless retaliation representation, and decomposed Bellman learning targets.
- **Empirical topology recovery.** FactorDRQN achieves 92.2% cell-level regime agreement (59/64 cells) with the tabular reference topology across the primary structural parameter grid, with residual disagreement concentrated near the Transition–Escalation boundary at high retaliation capacity, and 90.6% agreement under a second independent perturbation axis, confirming that topology recovery generalizes across perturbation dimensions.

II. RELATED WORK

A. Recurrent and Multi-Agent Reinforcement Learning

Deep Q-Networks [1] established end-to-end value learning from high-dimensional inputs, and Hausknecht and Stone [2] extended this to partially observable settings by replacing the first fully-connected layer with a recurrent unit, producing the Deep Recurrent Q-Network (DRQN). The gated recurrent unit [6] provides the recurrent backbone used in subsequent work, including our own, due to its favorable trade-off between memory capacity and training stability relative to LSTM. Value overestimation in deep Q-learning is addressed by Double DQN [3], which we adopt in both the tabular reference system and the recurrent architecture.

In multi-agent settings, independent Q-learning [11] and its convergence properties [12] established the foundational tension between decentralized execution and coordinated value learning. QMIX [8] introduced monotonic value function factorization to decompose joint action-value estimates across agents while maintaining centralized training, demonstrating

that structured decomposition of learning pathways is a principled design choice rather than an ad hoc approximation. MADDPG [10] and COMA [9] further established credit assignment as a central challenge in cooperative and mixed-motive multi-agent systems.

What this body of work does not address is the effect of recurrent architecture design on the qualitative structure of emergent strategic behavior. Evaluation in recurrent MARL has focused primarily on episodic return and task success rate, without characterizing whether learned policies recover the regime topology of the underlying strategic environment.

B. Reinforcement Learning in Strategic and Economic Systems

Reinforcement learning has been applied to strategic economic settings with increasing sophistication. Tesauro and Kephart [14] demonstrated that Q-learning agents in competitive pricing environments converge to tacit collusion under certain conditions, establishing that adaptive agents can produce qualitatively distinct equilibrium outcomes depending on environment structure. Calvano et al. [13] extended this result in a rigorous economic framework, showing that algorithmic pricing agents reliably discover supracompetitive equilibria without explicit coordination. Waltman and Kaymak [16] studied Q-learning dynamics in Cournot oligopoly models, finding that convergence behavior depends sensitively on exploration and learning rate parameters. The AI Economist [15] scales multi-agent RL to two-level policy design in simulated economies, demonstrating that deep MARL can recover interpretable policy structure in complex socio-economic environments.

These approaches evaluate agent behavior primarily through economic outcomes: prices, quantities, welfare, or tax rates. None examines whether the learned system recovers a reference phase topology across structural parameters, or how recurrent architecture design affects the qualitative geometry of emergent strategic regimes.

C. Phase Structure, Regime Topology, and Evaluation

The study of qualitative regime structure in dynamical systems has a long history in nonlinear dynamics and complex systems theory [18], where phase diagrams characterize how system behavior changes qualitatively as structural parameters vary. Recent work has brought this perspective into multi-agent reinforcement learning: Yamaguchi [17] constructs phase maps for independent Q-learning across environment size and agent density, identifying three distinct regimes separated by an instability ridge, and shows that small asymmetries between agents are a necessary driver of regime structure. This work establishes that MARL systems can exhibit coherent phase topology governed by structural parameters, rather than a smooth continuum of performance.

A parallel line of work has questioned whether scalar reward is a sufficient evaluation criterion for learned behavior. Amodei et al. [19] identify reward misspecification as a central failure mode in deployed RL systems, and Pan et al. [20] demonstrate empirically that agents optimizing misspecified rewards can

achieve high returns while developing qualitatively misaligned behavior. These results motivate evaluation frameworks that go beyond episodic return.

Our work extends this direction to recurrent multi-agent systems in asymmetric strategic environments, proposing phase-topology recovery as an evaluation criterion and identifying the architectural conditions under which recurrent function approximation preserves reference regime structure.

III. SOCIO-ECONOMIC ENVIRONMENT

A. Game Structure

We study a two-country repeated interaction in which a leader agent sets import tariffs and a follower agent responds with currency depreciation and export retaliation. The two agents have asymmetric instruments and asymmetric payoff sensitivities, producing a strategic environment in which the leader holds a structural capacity advantage but faces threshold-dependent retaliation from the follower. Both agents observe the current state and act within each period, but their instruments operate on different timescales and enter the opponent's payoff through distinct channels. No closed-form equilibrium is assumed; agents discover policies through repeated interaction.

The leader selects a tariff $\tau \in \{0, \dots, \tau_{\max}\}$ discretized into six bins. The follower selects a depreciation level $d \in \{0, \dots, d_{\max}\}$ across six bins and a retaliation intensity $\rho \in \{0, \dots, \rho_{\max}\}$ across five bins, with the two instruments serving distinct strategic roles.

B. Trade Dynamics

The economic state consists of import volume M_t , export volume X_t , and exchange rate E_t , each evolving by mean reversion toward instrument-dependent targets:

$$M_t = (1 - \kappa)M_{t-1} + \kappa \cdot M^*, \quad M^* = M_0(1 + \tau)^{-\varepsilon_d} \quad (1)$$

$$X_t = (1 - \lambda)X_{t-1} + \lambda \cdot X^*, \quad X^* = X_0(1 + d)^{\varepsilon_s} \quad (2)$$

$$E_t = (1 - \eta)E_{t-1} + \eta \cdot E^*, \quad E^* = E_0(1 + \rho)^{-\varepsilon_E} \quad (3)$$

where κ, λ, η are mean-reversion speeds and $\varepsilon_d, \varepsilon_s, \varepsilon_E$ are import demand, export supply, and leader export elasticities. The mean-reversion structure introduces delayed feedback: current instrument choices affect payoffs over multiple periods, creating the temporal credit assignment challenge that motivates recurrent architectures.

C. Payoff Structure

Period payoffs for the leader (L) and follower (F) are:

$$L = \tau M - \frac{1}{2}\tau(M_0 - M) - c_L \tau^2 - \psi_E \max(0, E_0 - E) - \xi \frac{\rho}{\rho_{\max}}(\tau M) \quad (4)$$

$$F = (X - X_0) - \phi_F d I_{F0} - c_F d^2 - c_\rho \rho^2 + \beta_E \cdot \tau \max(0, E_0 - E) \quad (5)$$

The leader gains from tariff revenue τM but incurs quadratic instrument costs and an exchange-rate penalty $\psi_E \max(0, E_0 - E)$ when currency appreciation erodes competitiveness. The diplomatic cost term $\xi(\rho/\rho_{\max})(\tau M)$ reduces leader payoff as follower retaliation intensity rises. The

TABLE I
STRUCTURAL CALIBRATIONS: LF, CENTER, AND FF.

Parameter	LF	CENTER	FF
ε_d (demand elast.)	1.30	1.42	1.55
ε_s (supply elast.)	0.85	1.00	1.20
ε_E (export elast.)	1.00	1.25	1.50
Import adj. speed (κ)	0.20	0.60	0.60
Export adj. speed (λ)	0.40	0.55	0.55
Exch. rate adj. speed (η)	0.30	0.55	0.55
Tariff ceiling ($\tau_{\max}$)	0.35	0.28	0.25
Depreciation ceiling ($d_{\max}$)	0.18	0.22	0.25
Retaliation ceiling ($\rho_{\max}$)	0.20	0.27	0.35
Exch. rate sensitivity (ψ_E)	0.90	1.15	1.40
Discount factor (δ)	0.98	0.98	0.98

LF: leader-favoring; FF: follower-favoring. Elasticity ranges are consistent with empirical estimates from recent trade conflict studies [21], [22].

follower gains from export expansion ($X - X_0$) but bears a linear inflationary cost $\phi_F d I_{F0}$ scaled by baseline import exposure I_{F0} , quadratic depreciation costs, and quadratic retaliation costs. The retaliation instrument ρ does not directly increase follower export revenue; its role is purely strategic, operating through its effect on leader payoffs and the subsequent leader response. The spite term $\beta_E \cdot \tau \max(0, E_0 - E)$ enters the follower payoff and governs the intensity of exchange-rate conflict; its magnitude is one axis of the structural parameter grid. In the baseline calibration $I_{F0} = M_0 = 450$, so that payoff differences are driven by strategic interactions rather than asymmetric import exposure.

D. Calibration and Structural Parameter Grid

Table I gives the baseline calibration used throughout. Parameter ranges are informed by empirical estimates of trade elasticities [21], [22]. Three representative operating calibrations serve as consistent cross-method reference points: leader-favoring (LF), balanced (CENTER), and follower-favoring (FF), corresponding to progressively increasing retaliation advantage, with $\rho_{\max} = 0.20, 0.28, 0.35$ respectively and all other parameters at baseline.

Two structural parameter axes are swept independently across eight levels each, producing $8 \times 8 = 64$ configurations per axis:

- **Exchange-rate conflict intensity** ($\beta_E \in [0, 1]$): the primary topology recovery benchmark, varying follower sensitivity to exchange-rate displacement under fixed diplomatic cost ($\xi = 0$).
- **Diplomatic cost perturbation** ($\xi \in [0, 3.0]$): the robustness axis, varying the leader's diplomatic penalty under fixed exchange-rate conflict intensity ($\beta_E = 0.5$).

Phase topology is evaluated independently across each axis, with the exchange-rate conflict sweep serving as the primary benchmark and the diplomatic cost sweep providing a second independent test of topology stability.

E. Tabular Reference System

We establish a tabular double Q-learning baseline as a reference phase topology for evaluating recurrent multi-agent architectures under identical economic dynamics. The leader

uses standard single Q-learning; the follower uses double Q-learning [4] over the joint (d, ρ) action space to reduce over-estimation bias across instruments. Training runs for 600,000 steps per seed, with exploration decaying from $\varepsilon = 0.15$ to $\varepsilon = 0.02$ over the training horizon. For each of the 64 parameter configurations per axis, five independent seeds are run and the resulting regime assignments are aggregated by majority vote to produce the reference topology.

This system serves as a low-dimensional interpretable reference learner. Its purpose is not to characterize optimal economic policy but to establish a phase topology against which the topology-preservation properties of recurrent architectures can be assessed. Reward-equivalent policies can nevertheless produce qualitatively different regime organization across structural parameter grids, motivating topology rather than return as the primary evaluation criterion. The reference topology exhibits three regimes defined in Section V, with consistent boundary geometry across seeds under this calibration. Both the tabular reference and the recurrent architecture use identical payoff-first step ordering, ensuring that temporal credit assignment conditions are not a confound in the topology comparison.

IV. RECURRENT MULTI-AGENT ARCHITECTURE

Both agents are implemented as recurrent deep Q-networks with dueling architecture [5] and Double DQN targets [3], trained via temporal-difference learning [7]. The leader and follower each maintain separate online and target networks; target networks are updated via soft interpolation ($\tau_{\text{soft}} = 0.015$; here τ_{soft} denotes the soft update coefficient, distinct from the tariff level τ used throughout) at every step, with periodic hard synchronization every 80 episodes to suppress Q-value inflation. Experience is stored in sequence replay buffers [2] of capacity 300 episodes (leader) and 100 episodes (follower), sampled with burn-in of 10 steps and unroll of 32 steps. Training runs for 1,200 episodes of 256 steps each. Exploration follows asymmetric ε -greedy schedules: the leader decays from $\varepsilon = 1.0$ to 0.05 over 120,000 steps ($\approx$ episode 469); the follower decays over 60,000 steps ($\approx$ episode 234), allowing the follower to converge before the leader commits to a tariff policy. For the first 100 episodes, the follower cycles through all tariff bins during training to ensure the retaliation pathway encounters the full tariff range before the leader policy hardens. Full training configuration is given in Table II.

A. Naive DRQN Failure Modes

We identify three compounding architectural pathologies in the naive recurrent architecture (A1) that prevent phase-topology recovery.

Reward-scale domination. In the naive architecture the follower receives a single shared reward equal to total follower payoff F , combining a depreciation-driven export gain of approximately 30 per step at convergence with a retaliation spite component of approximately 1.75 per step at high τ . The Q-function primarily learns variation associated with d , while

TABLE II
FACTORDRQN TRAINING CONFIGURATION.

Parameter	Value
Episodes	1,200
Episode length	256 steps
Burn-in / unroll	10 / 32 steps
Batch size	32
Learning rate (both agents)	3×10^{-4}
Discount factor (γ)	0.98
Leader replay capacity	300 episodes
Follower replay capacity	100 episodes
Leader ε decay	$1.0 \rightarrow 0.05$, 120,000 steps
Follower ε decay	$1.0 \rightarrow 0.05$, 60,000 steps
Follower curriculum	100 episodes
Soft update (τ_{soft})	0.015
Hard sync interval	80 episodes
Depreciation reward scale (Δ_d)	30
Retaliation reward scale (Δ_ρ)	1.0
τ input noise (σ)	0.02 (training only)

variation associated with ρ appears as a small perturbation. Because the Huber loss gradient is dominated by the larger depreciation signal, retaliation gradients fall below the effective noise floor and the agent remains approximately indifferent across retaliation levels.

Credit assignment collapse. With a joint 30-action space ($6_d \times 5_\rho$), the Q-function must partition value across all action combinations. Because reward variation is dominated by d , the five ρ -bins within each d -slice receive near-identical Q-values under greedy action selection, and the agent cannot distinguish retaliation levels given any fixed depreciation choice. The result is that ρ acts as a free variable in the learned policy: any retaliation level is consistent with the approximately optimal depreciation choice, producing regime boundaries that reflect d -learning but not strategic retaliation thresholds.

Recurrent overwrite of the retaliation signal. In the naive recurrent model, the retaliation signal is forced through the same hidden state used to track evolving economic flows, even though retaliation is primarily tied to current tariff provocation. The GRU hidden state is updated at every step by high-frequency economic variables (M, X, E), which continuously displace the strategic posture information relevant to ρ . This architectural mismatch means the retaliation credit signal is structurally diluted before it can be reinforced, independent of training duration or learning rate.

Panel (b) of Fig. 3 shows the resulting phase topology: the non-monotone zero-payoff boundary contour and isolated Deterrence region at high retaliation and tariff capacity are consistent with unstable value estimates produced by reward-scale domination, joint-action credit dilution, and recurrent overwrite operating simultaneously.

B. FactorDRQN: Factorized Follower Architecture

FactorDRQN addresses each pathology through three targeted architectural choices, illustrated in Fig. 1.

Reward decomposition. The environment exposes separate per-step signals r_d and r_ρ for the depreciation and retaliation components of follower payoff. Separate Bellman targets are

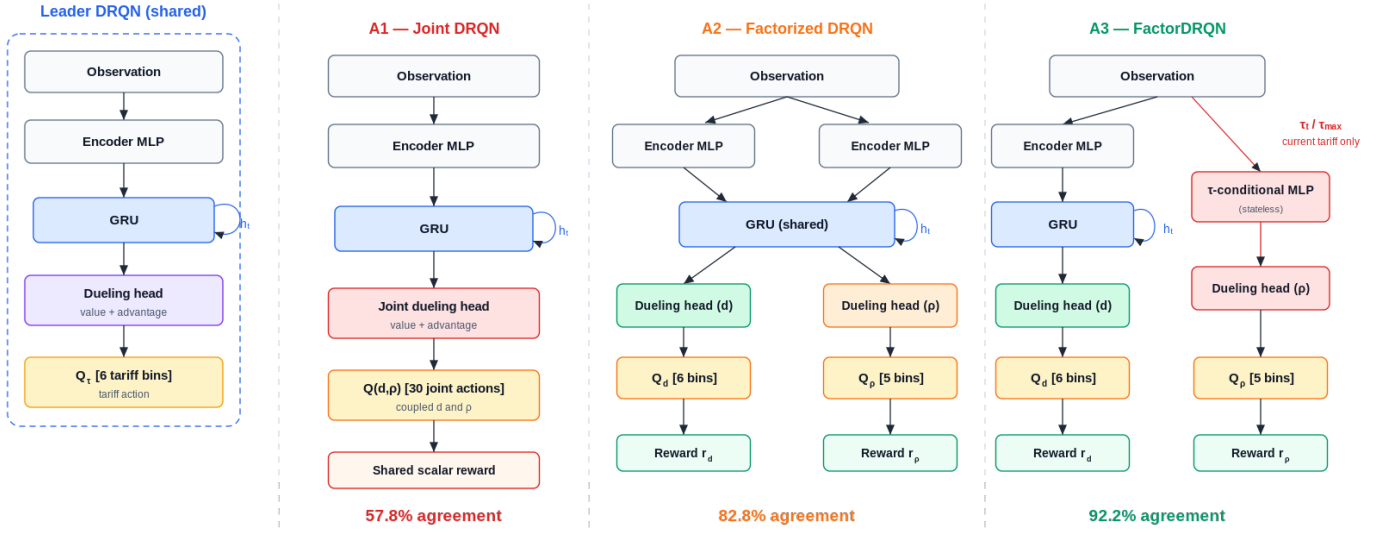


Fig. 1. Leader-follower DRQN architecture comparison across progressive architectural conditions. The leader agent (left) is identical in all conditions: a GRU trunk with a dueling Q-head producing Q_τ over six tariff bins. Follower architectures vary. **A1** (Naive DRQN): a single GRU processes the full observation and feeds a joint dueling head producing $Q(d, \rho)$ over 30 coupled actions ($6d \times 5p$); the agent receives a single shared scalar reward. **A2** (Factorized DRQN): both the depreciation and retaliation pathways share a single GRU trunk, then split into separate dueling heads producing Q_d and Q_p ; rewards are decomposed into per-head Bellman targets. **A3** (FactorDRQN, proposed): the depreciation pathway retains the GRU trunk; the retaliation pathway bypasses the GRU entirely, receiving only the current normalized tariff $\tau_t/\tau_{\max}$ as input to a stateless τ -conditional MLP. Cell-level topology agreement with the tabular reference is shown for each condition (A1: 57.8%, A2: 82.8%, A3: 92.2%).

computed for each head:

$$y_d = r_d/\Delta_d + \gamma \max_{d'} Q_d(s', d') \quad (6)$$

$$y_p = r_p/\Delta_p + \gamma \max_{p'} Q_p(s', p') \quad (7)$$

where $\Delta_d = 30$ normalizes the depreciation reward to unit scale and $\Delta_p = 1$ preserves the retaliation signal. Reward clipping to $[-3, 3]$ for r_d/Δ_d and $[-0.1, 0.1]$ for r_p prevents outlier transitions from dominating bootstrap targets, eliminating gradient interference between the two heads.

Factorized action heads. The follower maintains two independent Q-heads over separate action spaces: Q_d over six depreciation bins and Q_p over five retaliation bins. Actions are selected independently as $\arg \max Q_d$ and $\arg \max Q_p$, removing the credit assignment coupling that renders ρ a free variable in the joint policy.

τ -conditional retaliation pathway. The Q_p head receives only the current normalized tariff $\tau_t/\tau_{\max}$ as input, bypassing the GRU trunk entirely. This stateless MLP, $\text{Linear}(1 \rightarrow 64)$, ReLU , $\text{Linear}(64 \rightarrow 64)$, dueling head, produces Q_p as a pure function of current tariff provocation, preventing economic-flow updates from overwriting the retaliation signal. This separation encodes an environment-specific inductive bias: in this environment, retaliation threshold behavior depends primarily on current tariff intensity rather than accumulated history, and recurrent entanglement suppresses this threshold structure. During training, additive Gaussian noise ($\sigma = 0.02$) is injected into the τ input to regularize Q-values against drift when the leader converges to a fixed tariff level; this noise is suppressed at evaluation time.

Progressive topology recovery. Table III and Fig. 3 show the contribution of each architectural condition. Factorization

TABLE III
PROGRESSIVE TOPOLOGY RECOVERY.

Architecture	Agreement	Δ
A1 Naive DRQN (joint action, shared reward)	57.8%	—
A2 Factorized DRQN (GRU $\rightarrow \rho$ -head)	82.8%	+25.0 pp
A3 FactorDRQN (τ -conditional ρ -head)	92.2%	+9.4 pp

64 cells, 5-seed majority vote. Reference: tabular double Q-learning.

alone (A2) recovers 82.8% cell-level topology agreement with the tabular reference, a gain of 25.0 percentage points over the naive baseline. Adding the τ -conditional retaliation pathway (A3) provides a further 9.4 percentage point improvement, bringing total agreement to 92.2%. The five remaining disagreements are concentrated at the Transition–Escalation boundary at high $\rho_{\max}$, consistent with stochastic policy variance near the regime threshold rather than systematic architectural failure.

V. PHASE-TOPOLOGY EVALUATION AND METRICS

Regime classification. Each cell in the 8×8 structural parameter grid is assigned a regime by applying a payoff-primary threshold to the mean leader payoff computed over the evaluation window (episodes 900–1200 of 1,200 total). For each episode, the per-step leader reward is averaged over all 256 steps; these episode means are then averaged across all evaluation episodes to produce a single scalar per seed. A cell is assigned Deterrence if $\bar{L} > +15$, Escalation if $\bar{L} < -15$, and Transition otherwise. Leader payoff integrates tariff revenue, instrument costs, exchange-rate penalties, and retaliation response into a single scalar, making it the most informative single indicator of strategic regime across structural parameter configurations. The ± 15 thresholds were chosen to produce

balanced regime coverage across the structural parameter grid; results are qualitatively stable to moderate threshold variation.

Seed aggregation. Five independent seeds are run per configuration. Regime assignment follows strict majority vote: a regime is assigned if it appears in at least three of five seeds. In the absence of a strict majority, the cell is classified by applying the ± 15 threshold to the cross-seed mean leader payoff. No ties required this fallback in the results reported here.

Cell-level topology agreement. The primary evaluation metric is cell-level topology agreement between the recurrent architecture and the tabular reference topology:

$$\text{Agreement} = \frac{1}{64} \sum_{i=1}^{64} \mathbf{1}[\hat{r}_i = r_i^*] \quad (8)$$

where $\hat{r}_i$ is the majority-vote regime assignment for cell i under the recurrent architecture and r_i^* is the corresponding tabular reference assignment. An agreement of 1.0 indicates full topology recovery; the tabular reference agrees with itself at 1.0 by construction across seeds.

Boundary geometry. The zero-payoff boundary contour, shown as a black curve in Fig. 3, marks the transitions between Deterrence, Transition, and Escalation regions across the parameter grid. Boundary geometry is evaluated by contour monotonicity: a well-recovered topology produces a smooth, monotone boundary separating regimes; a distorted topology produces a non-monotone or fragmented contour. This qualitative criterion is appropriate for an 8×8 discrete grid, where fine-grained distance metrics such as Hausdorff distance are not meaningful at cell resolution. Disagreements between architectures are reported both as cell counts and by their location relative to the boundary.

VI. RESULTS

A. Naive Topology Collapse

Panel (b) of Fig. 3 shows the phase topology produced by the naive recurrent architecture (A1) across the retaliation capacity vs tariff capacity grid. The reference topology in panel (a) exhibits a smooth, monotone boundary separating Deterrence, Transition, and Escalation regions, with the LF calibration anchor firmly in the Deterrence region and the FF anchor near the Escalation boundary.

The naive architecture distorts this topology in three observable ways. First, the zero-payoff boundary contour is non-monotone: the boundary reverses direction at high $\rho_{\max}$, producing a jagged contour that does not reflect the monotone strategic trade-off of the reference. Second, an isolated Deterrence island appears at $\rho_{\max} \approx 0.36\text{--}0.43$, $\tau_{\max} \approx 0.35\text{--}0.37$, a configuration where the tabular reference assigns Escalation. Third, the Transition band is distorted in width and position relative to the reference, with irregular incursions of Deterrence into the Transition region at mid-range $\rho_{\max}$. These distortions are consistent with the three failure modes identified in Section IV-A: reward-scale domination suppresses retaliation learning, credit assignment collapse renders ρ a free variable, and recurrent overwrite prevents stable threshold

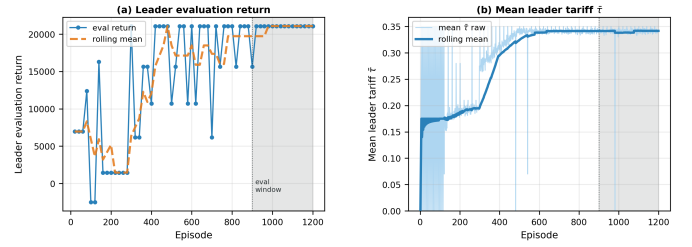


Fig. 2. FactorDRQN training stabilization (single representative run, LF calibration). Left: leader evaluation return rises from approximately 7,000 to 21,000 by episode 700, with a clear inflection at episode ≈ 470 when the leader ε reaches its floor. Right: mean leader tariff $\bar{\tau}$ increases from 0.17 to 0.35 between episodes 200–600, then stabilizes, marking commitment to a high-tariff policy. The follower evaluation return (not shown) stabilizes by episode ≈ 300 . The evaluation window (episodes 900–1200, shaded) falls entirely within the stable greedy phase for both agents.

formation. Cell-level topology agreement with the tabular reference is 57.8% (37/64 cells).

B. Training Stabilization

Fig. 2 shows training stabilization for FactorDRQN under the LF calibration. The follower converges by episode ≈ 300 , well before the evaluation window opens at episode 900. The leader exhibits a clear inflection at episode ≈ 470 , coinciding with the ε floor, after which the mean tariff $\bar{\tau}$ rises from 0.17 to 0.35 and stabilizes by episode ≈ 600 . The evaluation window at episodes 900–1200 therefore captures fully converged greedy policies for both agents, with approximately 300 greedy episodes of stable interaction preceding evaluation. These diagnostics confirm that topology evaluation is not confounded by incomplete training.

C. Topology Recovery

Fig. 3 shows the four-panel topology comparison across the retaliation capacity vs tariff capacity grid. Table III reports cell-level agreement with the tabular reference for each architectural condition.

The progression from A1 to A3 is monotone and interpretable. The largest single gain – 25.0 percentage points – arises from pathway factorization alone (A1 to A2). Separating the depreciation and retaliation Q-heads eliminates the credit assignment coupling that renders ρ a free variable, recovering the large-scale three-regime structure and restoring approximate boundary monotonicity. The remaining gain – 9.4 percentage points – arises from adding the τ -conditional retaliation pathway (A2 to A3), which refines local boundary fidelity by preventing economic-flow updates from overwriting the retaliation threshold signal.

FactorDRQN achieves 92.2% cell-level agreement (59/64 cells) with the tabular reference. The five disagreeing cells are concentrated at the Transition–Escalation boundary at high $\rho_{\max}$, where stochastic seed variance near the regime threshold produces occasional misclassification. No systematic structural failure is evident: the boundary contour in panel (d) is smooth and monotone, regime proportions closely match panel (a), and the LF, CENTER, and FF calibration anchors fall in their reference regimes.

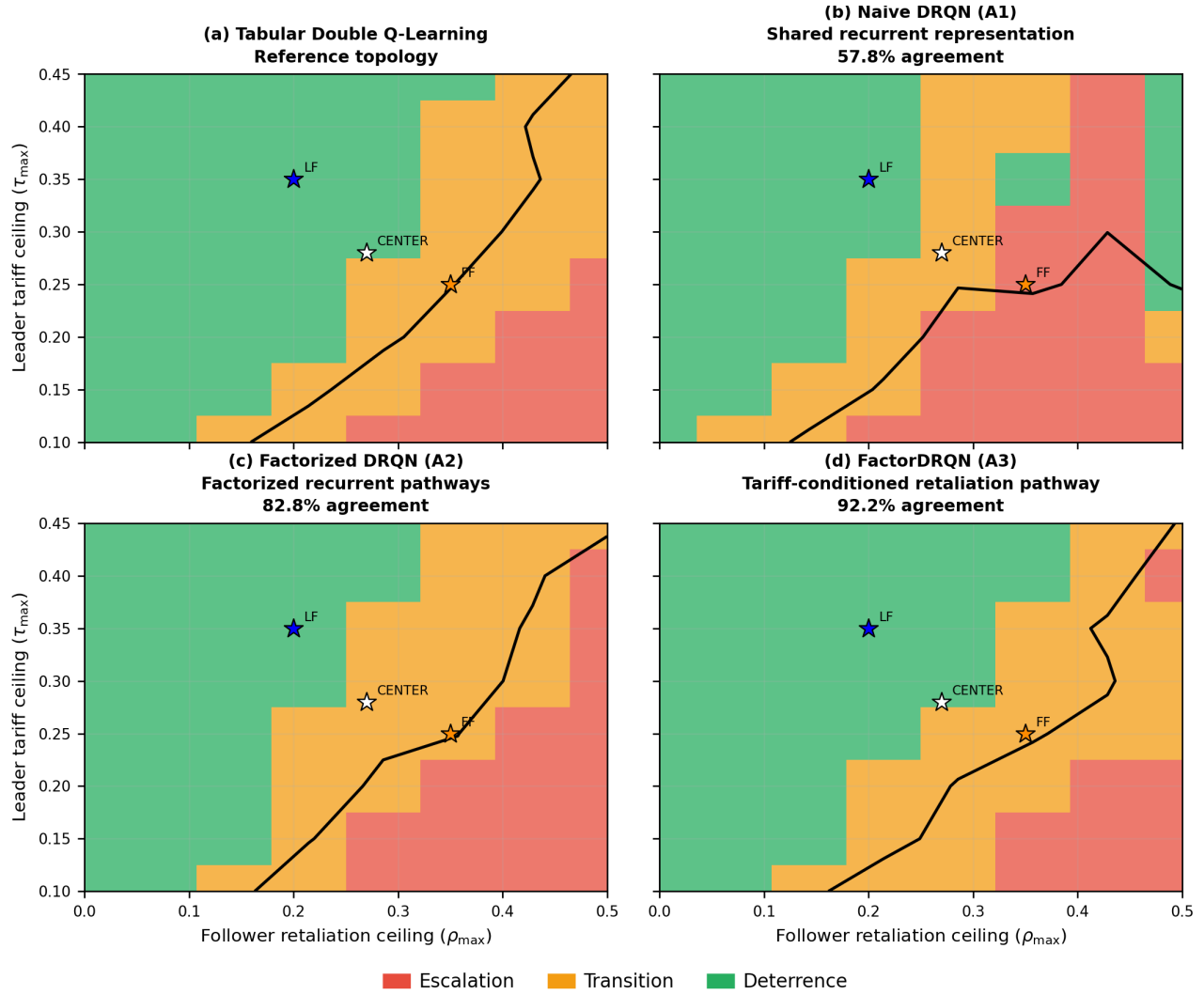


Fig. 3. Recovery of the reference phase topology under progressively imposed architectural conditions. **(a)** Tabular double Q-learning reference topology with coherent Deterrence, Transition, and Escalation regions and stable boundary geometry. **(b)** Naive DRQN (A1, 57.8% agreement): non-monotone boundary contour, isolated Deterrence island at high capacity, and Transition-band distortion consistent with reward-scale domination, credit assignment collapse, and recurrent overwrite. **(c)** Factorized DRQN (A2, 82.8% agreement): large-scale strategic topology recovered through pathway factorization; local boundary geometry remains rougher than the reference at high $\rho_{\max}$. **(d)** FactorDRQN (A3, 92.2% agreement): smooth monotone boundary contour, regime proportions closely matching the reference, with residual disagreement concentrated near the Transition–Escalation boundary at high retaliation capacity. Stars mark the LF (filled), CENTER, and FF (open) calibration anchors.

D. Robustness Under Diplomatic Cost Perturbation

Fig. 4 shows the FactorDRQN phase topology across the diplomatic cost perturbation axis ($\xi \in [0, 3.0]$, $\beta_E = 0.5$ fixed). FactorDRQN achieves 90.6% cell-level agreement on this axis, compared to 92.2% on the primary structural axis, confirming that topology recovery is not grid-specific and generalizes across independent perturbation dimensions.

The macroscopic three-regime structure is preserved: Deterrence persists at low ξ and low $\rho_{\max}$, Escalation dominates at high $\rho_{\max}$, and a monotone Transition band separates the two regions across the full perturbation range.

The primary structural effect of increasing ξ is a contraction of the Deterrence region in the $\rho_{\max}$ dimension. At $\xi = 0$, the Deterrence–Transition boundary sits at $\rho_{\max} \approx 0.20$; by $\xi = 1.5$, it has contracted to $\rho_{\max} \approx 0.07$. This is structurally coherent: diplomatic costs increase the leader’s penalty for

high-tariff policies under retaliation, reducing the parameter range in which Deterrence is sustainable.

The five disagreeing cells are concentrated near regime boundaries, consistent with the boundary variance pattern observed on the primary axis. No systematic structural failure is evident across the perturbation range.

VII. DISCUSSION AND LIMITATIONS

Instrument-temporal-structure principle. The central architectural lesson from this work is that instruments with different temporal dependencies require separated learning pathways. In the leader-follower tariff environment, the depreciation instrument is history-dependent: its value accumulates over multiple periods through mean-reversion dynamics, making recurrent representation appropriate. The retaliation instrument is near-Markov: its strategic value depends pri-

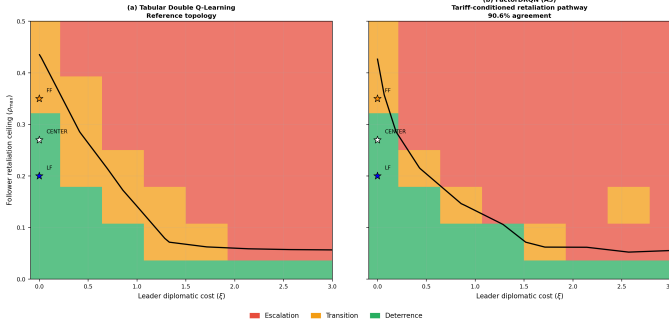


Fig. 4. FactorDRQN phase topology under diplomatic cost perturbation (ξ axis, $\beta_E = 0.5$ fixed). FactorDRQN achieves 90.6% cell-level agreement on this axis, compared to 92.2% on the primary structural axis, confirming that topology recovery generalizes across independent perturbation dimensions. As ξ increases from 0 to 3.0, the Deterrence region contracts monotonically in the $\rho_{\max}$ dimension, reflecting increasing diplomatic costs to the leader under high-tariff policies.

marily on current tariff provocation rather than accumulated history, making recurrent entanglement harmful. A shared GRU state conflates these two timescales, destroying the threshold structure of the reactive instrument regardless of training duration. Architectural separation restores it. This observation generalizes cautiously: in environments where instruments have heterogeneous temporal dependencies, pathway separation may be important for topology-preserving learning, though the claim is limited to the environment class studied here. **Reward decomposition and gradient interference.** The $\Delta_d = 30$ reward normalization is environment-specific, but the underlying problem is general: in any multi-instrument agent where instruments produce payoffs of substantially different magnitudes, gradient interference across a shared Q-function will suppress learning of the lower-magnitude instrument. Decomposed Bellman targets with per-instrument scaling is a straightforward architectural response that does not require environment-specific tuning beyond identifying the reward-scale ratio, and may transfer to other multi-instrument strategic settings where instrument payoffs are heterogeneous. **Phase-topology evaluation.** Scalar return metrics are insufficient for evaluating strategic multi-agent systems when qualitatively distinct equilibria yield similar episodic rewards. The cell-level topology agreement metric operationalizes a structural evaluation criterion for parameter-sweep environments: it directly measures whether the learned system recovers the regime organization of a reference learner rather than merely matching its average performance. This framework is applicable to any environment where a reference topology can be established and structural parameters can be swept systematically; the boundary geometry criterion provides a complementary qualitative diagnostic that scalar agreement does not capture.

Limitations. Three limitations bound the scope of the reported results.

First, the τ -conditional retaliation pathway is sensitive to the spite parameter β_E . Preliminary results at $\beta_E = 0$ suggest topology collapse: the retaliation gradient vanishes, the τ -conditional pathway produces near-uniform Q_ρ across retaliation levels, and the learned topology collapses to uni-

versal Deterrence. The $\beta_E = 0.5$ configuration restores topology recovery at 92.2% agreement, but the sensitivity of the pathway across the intermediate range $(0, 0.5)$ has not been characterized. Whether a minimum spite threshold exists below which the τ -conditional pathway degrades is an open empirical question.

Second, FactorDRQN’s Transition band is systematically slightly wider than the tabular reference near regime boundaries, visible in both the primary grid and the diplomatic cost perturbation axis. This residual bias – the recurrent architecture assigns Transition where the tabular reference assigns Escalation at boundary cells – likely reflects differences in value function smoothness between tabular and function-approximation learners rather than architectural failure. It is a known residual discrepancy rather than a correctable error.

Third, all results are established for a single environment class. The architecture conditions identified here are associated with topology recovery in asymmetric leader-follower conflict with factorized instruments and sparse retaliation signals. Generalization to other strategic multi-agent environments with different instrument structures, symmetric roles, or continuous action spaces is not established by this work.

VIII. CONCLUSION

We studied whether recurrent multi-agent learning systems can recover the phase topology of a two-country leader-follower tariff conflict under function approximation, using cell-level regime agreement with a tabular reference topology as the primary evaluation criterion. Naive recurrent deep Q-networks fail to do so, producing distorted topologies through three systematic pathologies: reward-scale domination, credit assignment collapse across factorized instruments, and recurrent overwrite of sparse strategic signals. The Factorized DRQN (FactorDRQN), which addresses each pathology through decomposed Bellman targets, separated action heads, and a τ -conditional stateless retaliation pathway, recovers 92.2% of the reference topology across an 8×8 structural parameter grid, with residual disagreement concentrated near the Transition–Escalation boundary at high retaliation capacity, and 90.6% agreement under a second independent perturbation axis. The monotone progression from 57.8% to 82.8% to 92.2% agreement across architectural conditions confirms that pathway factorization is the primary driver of topology recovery, with τ -conditioning providing targeted refinement at the regime boundary. These results contribute a phase-topology evaluation framework, a failure mode taxonomy for recurrent multi-agent systems in strategic economic environments, and a set of architecture conditions associated with stable topology recovery in this class of environments.

ACKNOWLEDGMENTS

The authors thank ASDRP for providing the research environment and mentorship that supported this work.

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